CosineSimilarity#

https://pytorch.org/docs/stable/generated/torch.nn.CosineSimilarity.html#tor

Description:

Returns cosine similarity between x1x_1x1 and x2x_2x2, computed along dim. dim (int, optional) – Dimension where cosine similarity is computed. Default: 1 eps (float, optional) – Small value to avoid division by zero. Default: 1e-8 Input1: (∗1,D,∗2)(\ast_1, D, \ast_2)(∗1,D,∗2) where D is at position dim Input2: (∗1,D,∗2)(\ast_1, D, \ast_2)(∗1,D,∗2), same number of dimensions as x1, matching x1 size at dimension dim, and broadcastable with x1 at other dimensions.

Function Signature

```
class torch.nn.CosineSimilarity(dim=1, eps=1e-08)[source]#
Parameters
Shape:
forward(x1, x2)[source]#
Return type
```

Returns:

Tensor

Code Examples

```
>>> input1 = torch.randn(100, 128)
>>> input2 = torch.randn(100, 128)
>>> cos = nn.CosineSimilarity(dim=1, eps=1e-6)
>>> output = cos(input1, input2)
```

Detailed Documentation

Documentation

Returns cosine similarity between x1x_1x1 and x2x_2x2, computed along dim.

dim (int, optional) – Dimension where cosine similarity is computed. Default: 1

eps (float, optional) – Small value to avoid division by zero. Default: 1e-8

Input1: (∗1,D,∗2)(\ast_1, D, \ast_2)(∗1,D,∗2) where D is at position dim

Input2: (∗1,D,∗2)(\ast_1, D, \ast_2)(∗1,D,∗2), same number of dimensions as x1, matching x1 size at dimension dim, and broadcastable with x1 at other dimensions.

Output: (∗1,∗2)(\ast_1, \ast_2)(∗1,∗2)

Runs the forward pass.